

Resurrection Musical Instruments



A Project Documentation

Presented to

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INVENTORY MANAGEMENT SYSTEM DOCUMENTATION

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INTRODUCTION

The Inventory Management System for **Resurrection Musical Instruments** is designed to manage products, stock levels, sales, and user information efficiently. The system helps organize inventory records, provide analytics, monitor product availability, and improve overall business operations.

As businesses continue to rely on digital systems, maintaining system security becomes increasingly important. Unauthorized access, data breaches, and inventory manipulation may lead to operational problems and financial loss. Therefore, this project focuses on improving both inventory management and system security to ensure data confidentiality, integrity, and availability.

The system provides user authentication, secure database management, and monitoring mechanisms to protect sensitive information while maintaining smooth and reliable operations.

OBJECTIVES

The objectives of the project are:

1. To develop an efficient inventory management system for managing products and stock.
 2. To improve user authentication and access control.
 3. To protect sensitive information using password encryption and secure communication.
 4. To provide organized product and sales management features.
 5. To enhance system monitoring and activity logging.
 6. To improve the overall efficiency of Resurrection Musical Instruments operations.
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METHODOLOGY

The development of the system follows several phases:

1. Planning and Requirement Gathering

The system requirements were analyzed to identify the needs of the business, including inventory tracking, product management, and user security.

2. System Design

The database structure, user interface, and system architecture were designed before implementation.

3. Development

The system was developed using web technologies such as PHP, JavaScript, HTML, CSS, and MySQL.

4. Security Implementation

Security measures such as encrypted passwords, secure login authentication, and validation mechanisms were implemented.

5. Testing

The system underwent functionality testing and security testing to ensure reliability and performance.

6. Deployment

After successful testing, the system was prepared for deployment and future maintenance.

TOOLS AND TECHNOLOGY

The following tools and technologies were used in the development of the system:

Category	Tools/Technology
	Programming Language PHP, JavaScript
Front-End	HTML, CSS, Bootstrap
Database	MySQL / MariaDB
Web Server	Apache (XAMPP)
Security Tools	OWASP ZAP
Version Control	GitHub
Protocols	HTTPS
Code Editor	Visual Studio Code

FRONT-END DOCUMENTATION

Front-End Overview

The front-end of the system provides a user-friendly interface for managing inventory, viewing products, and monitoring sales. The interface is designed to be responsive, organized, and easy to navigate.

Features

The system includes the following front-end features:

- User Login and Authentication
- Dashboard Overview
- Product Listing
- Add/Edit/Delete Products
- Search Functionality
- Inventory Monitoring
- User Profile Management
- Sales Tracking
- Reports and Logs
- Responsive Navigation

Pages and Components

Page/Component	Description
Login Page	Allows users to securely log in
Dashboard	Displays inventory statistics and system data
Product Management Page	Allows users to manage products
Inventory Page	Displays stock information

Search Bar	Allows quick product searching
User Profile Page	Manages user information
Reports Page	Displays sales and inventory reports
Settings Page	Allows customization of system settings

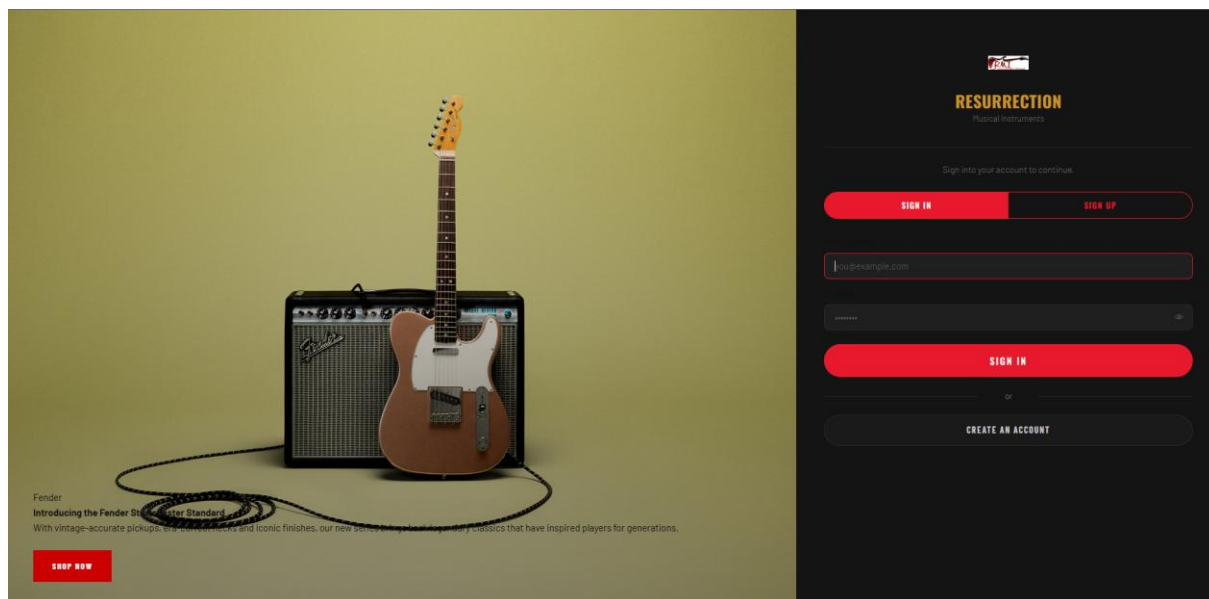
User Interface Design

The system interface uses a modern and responsive design with:

- Simple navigation menus
- Dashboard cards for quick statistics
- Organized tables for inventory records
- Search and filtering functionality
- Mobile-friendly layout

Sample UI Components

- Login Form
- Product Cards
- Inventory Tables
- Dashboard Analytics
- Sidebar Navigation





PRODUCTSAPPOINTMENTSMY ORDERSPROFILE

BARTCustomerLOGOUT

BROWSEOUR COLLECTIONPremium Instruments and accessories, ready for pickup

CART0

ALL (1)GUITAR (3)BASS (1)AMPLIFIER (1)STRINGS (2)PEDAL (1)ACCESSORY (1)DEFAULT

Search Instruments...

SEARCH

YOUR CART (0 ITEMS)

9 PRODUCTS FOUND



GUITAR

FENDER STRATOCASTER STANDARD

Classic SSS pickup configuration, maple neck

P45,000.00

ONLY 1 LEFT

CART



GUITAR

GIBSON LES PAUL STANDARD

Mahogany body, dual humbucker pickups

P85,000.00

ONLY 3 LEFT

CART



GUITAR

SQUIER CLASSIC VIBE 50S TELE

Vintage-style Telecaster, great for beginners

P18,500.00

IN STOCK

CART



BASS

VINTERA® III LATE '80S PRECISION BASS®

The Fender® Vintera III series delivers meticulously crafted vintage recre...

P25,000.00

IN STOCK

CART





YOUR CART IS EMPTY

TOTAL

P0.00

CHECKOUT →

[DASHBOARD](#)[ANALYTICS](#)[INVENTORY](#)[SALES / POS](#)[SERVICE](#)[APPOINTMENTS](#)[CUSTOMERS](#)[ADMIN](#)[LOGOUT](#)

DASHBOARD

WELCOME BACK, ADMIN – MONDAY, MAY 25, 2026

[MANAGE USERS](#)

TODAY'S REVENUE

P0.00

0 sales today

PENDING SERVICES

0

[View all →](#)

PENDING APPOINTMENTS

1

[View all →](#)

LOW STOCK ITEMS

2

[View Inventory →](#)

TOTAL PRODUCTS

9

[Manage →](#)

RECENT SALES

NEW SALE

#	CUSTOMER	PAYMENT	AMOUNT	TIME
#6	Daryl Aurelio	CASH	P45,000.00	08:12 PM
#5	Daryl Aurelio	CASH	P25,000.00	07:58 PM
#4	Walk-in	CASH	P450.00	06:34 PM
#3	Daryl Aurelio	CASH	P45,000.00	06:31 PM
#2	Daryl Aurelio	CASH	P4,500.00	01:28 PM
#1	Daryl Aurelio	CASH	P4,500.00	01:28 PM

LOW STOCK ALERT

VIEW ALL

Fender Stratocaster Standard
Guitar

2 LEFT

Gibson Les Paul Standard
Guitar

3 LEFT

Fender Frontman 100 Amp
Amplifier

5 LEFT

UPCOMING APPOINTMENTS

VIEW ALL

No upcoming appointments.

[DASHBOARD](#)[ANALYTICS](#)[INVENTORY](#)[SALES / POS](#)[SERVICE](#)[APPOINTMENTS](#)[CUSTOMERS](#)[ADMIN](#)[LOGOUT](#)

REPORTS

ANALYTICS

SALES PERFORMANCE & INSIGHTS

[LAST 7 DAYS](#)[LAST 30 DAYS](#)[LAST 90 DAYS](#)[THIS YEAR](#)

TOTAL REVENUE

P124,450.00

-- no prior data

TOTAL ORDERS

6

-- no prior data

AVG. ORDER VALUE

P20,741.67

-- no prior data

UNIQUE CUSTOMERS

1

registered accounts

REVENUE OVER TIME

Last 30 Days

BY PAYMENT METHOD

1 METHODS

Cash 100.0%

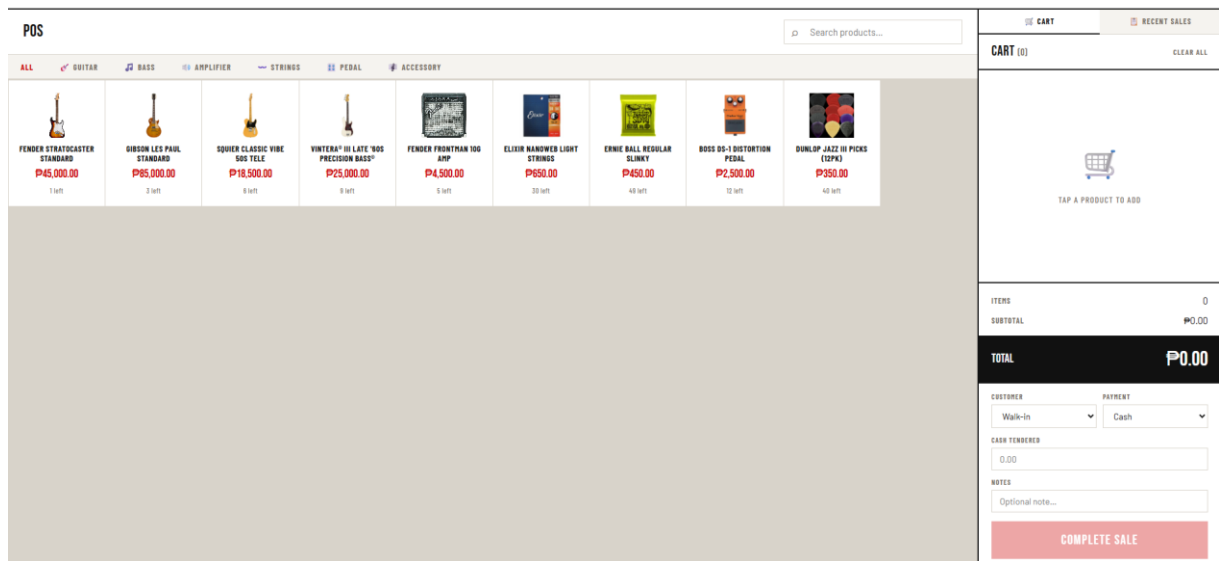
REVENUE BY CATEGORY

Guitar	P90,000.00 - 2 units
Bass	P25,000.00 - 1 units
Amplifier	P9,000.00 - 2 units

SALES BY HOUR

Darker = more sales activity

Peak hour: 1PM – 2 sales – P9,000.00



DATABASE DESIGN

Database Overview

The database for the **Resurrection Musical Instruments Inventory Management System** is designed to manage products, users, appointments, sales transactions, and service requests efficiently. The database uses relational table structures with primary keys and foreign keys to maintain data consistency and integrity.

The system supports:

- Product inventory management
- Customer and staff accounts
- Sales transactions
- Appointment scheduling
- Repair and maintenance service requests

The database was developed using **MariaDB/MySQL** and managed through phpMyAdmin.

ENTITY RELATIONSHIP DIAGRAM (ERD)

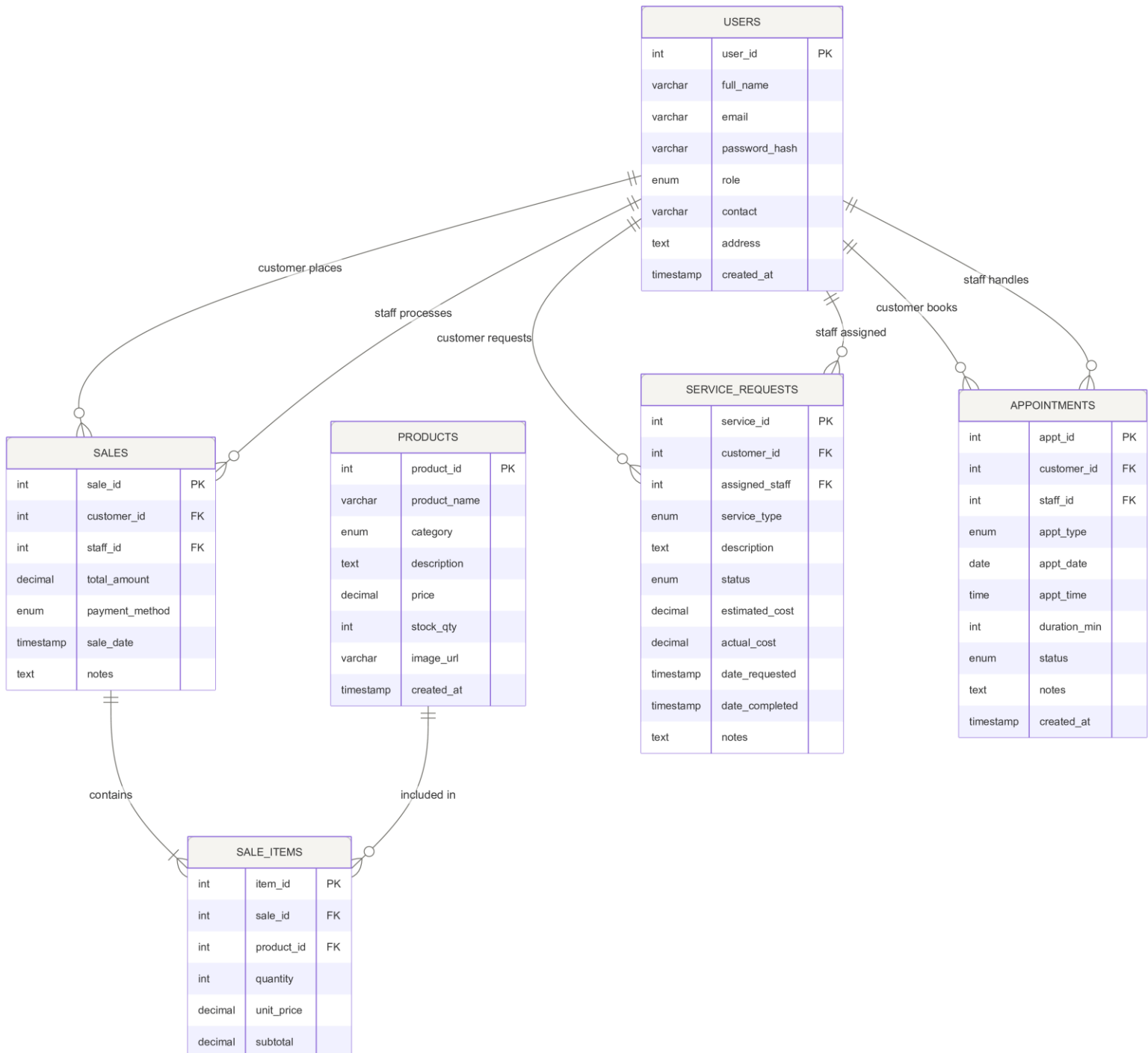
Main Entities

- Users

- Products
- Sales
- Sale Items
- Appointments
- Service Requests

Relationship Summary

- One user can create many appointments.
- One user can request many services.
- One sale can contain many sale items.
- One product can appear in many sale items.
- Staff users can manage appointments and services.



TABLES AND FIELDS

Table: Users

Stores all registered users including customers, staff, and administrators.

Field Name	Data Type	Description
user_id	INT	Primary Key
full_name	VARCHAR(100)	Full name of user
email	VARCHAR(150)	User email
password_hash	VARCHAR(255)	Encrypted password
role	ENUM	customer, staff, or admin
contact	VARCHAR(20)	Contact number
address	TEXT	User address
created_at	TIMESTAMP	Account creation date

Sample Roles

- Admin
- Staff
- Customer

Table: Products

Stores inventory products sold by the music store.

Field Name	Data Type	Description
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product_id	INT	Primary Key
product_name	VARCHAR(150)	Product name
category	ENUM	Product category
description	TEXT	Product description
price	DECIMAL(10,2)	Product price
stock_qty	INT	Current stock quantity
image_url	VARCHAR(255)	Main product image
image_url_2	VARCHAR(500)	Additional image
image_url_3	VARCHAR(500)	Additional image
image_url_4	VARCHAR(500)	Additional image
image_url_5	VARCHAR(500)	Additional image
created_at	TIMESTAMP	Product creation date

Product Categories

- Guitar
- Bass
- Amplifier
- Strings

- Pedal
- Accessory
- Other

Table: Appointments

Handles customer appointment scheduling.

Field Name	Data Type	Description
appt_id	INT	Primary Key
customer_id	INT	Customer reference
staff_id	INT	Assigned staff
appt_type	ENUM	Lesson, Repair, Consultation
appt_date	DATE	Appointment date
appt_time	TIME	Appointment time
duration_min	INT	Duration in minutes
status	ENUM	Appointment status
notes	TEXT	Additional notes
created_at	TIMESTAMP	Record creation date

Appointment Status

- Pending

- Confirmed
- Completed
- Cancelled

Table: Sales

Stores sales transaction information.

Field Name	Data Type	Description
sale_id	INT	Primary Key
customer_id	INT	Customer reference
staff_id	INT	Staff reference
total_amount	DECIMAL(10,2)	Total sale amount
payment_method	ENUM	Payment method
sale_date	TIMESTAMP	Sale transaction date
notes	TEXT	Additional notes

Payment Methods

- Cash
- GCash
- Card
- Bank Transfer

Table: Sale Items

Stores products included in each sale transaction.

Field Name	Data Type	Description
item_id	INT	Primary Key
sale_id	INT	Sale reference
product_id	INT	Product reference
quantity	INT	Product quantity
unit_price	DECIMAL(10,2)	Price per item
subtotal	DECIMAL(10,2)	Computed subtotal

The **subtotal** field is automatically generated using:

$$\text{subtotal} = \text{quantity} \times \text{unit_price}$$

Table: Service Requests

Stores customer repair and maintenance service requests.

Field Name	Data Type	Description
service_id	INT	Primary Key
customer_id	INT	Customer reference

assigned_staff	INT	Assigned technician/staff
service_type	ENUM	Type of service
description	TEXT	Service description
status	ENUM	Service status
estimated_cost	DECIMAL(10,2)	Estimated repair cost
actual_cost	DECIMAL(10,2)	Final repair cost
date_requested	TIMESTAMP	Request date
date_completed	TIMESTAMP	Completion date
notes	TEXT	Additional notes

Service Types

- Setup
- Repair
- String Replacement
- Electronics
- Cleaning
- Other

RELATIONSHIPS

The database uses relational mapping through foreign keys.

Relationship	Description
Users → Appointments	One customer can create many appointments
Users → Service Requests	One customer can request many services
Users → Sales	One customer can have many sales
Products → Sale Items	One product can appear in many sale items
Sales → Sale Items	One sale can contain multiple items
Staff → Appointments	One staff can manage many appointments
Staff → Service Requests	One staff can handle many services

DATABASE CONSTRAINTS

The following constraints are implemented:

Primary Keys

Used to uniquely identify each record.

Example:

PRIMARY KEY (product_id)

Foreign Keys

Used to maintain table relationships.

Example:

FOREIGN KEY (customer_id) REFERENCES users(user_id)

Unique Constraint

Prevents duplicate user emails.

UNIQUE KEY email (email)

NOT NULL Constraint

Ensures important fields cannot be empty.

Example:

product_name VARCHAR(150) NOT NULL

Auto Increment

Automatically generates IDs.

Example:

product_id INT AUTO_INCREMENT

SAMPLE SQL QUERIES

Display All Products

SELECT * FROM products;

Display Available Products

SELECT product_name, stock_qty

FROM products

```
WHERE stock_qty > 0;
```

View Sales with Product Information

```
SELECT s.sale_id, p.product_name, si.quantity, si.subtotal
```

```
FROM sales s
```

```
INNER JOIN sale_items si ON s.sale_id = si.sale_id
```

```
INNER JOIN products p ON si.product_id = p.product_id;
```

Display Pending Appointments

```
SELECT *
```

```
FROM appointments
```

```
WHERE status = 'Pending';
```

Display Service Requests

```
SELECT service_type, status, estimated_cost
```

```
FROM service_requests;
```

SECURITY IMPLEMENTATION

The system includes several security features:

- Password hashing using PHP password hashing functions
- User role-based access control
- Foreign key constraints for data consistency
- Input validation and sanitization
- Secure database relationships
- HTTPS-ready communication

REFLECTION

The database structure of the Resurrection Musical Instruments Inventory Management System provides an organized and secure method of managing inventory, users, appointments, and sales transactions. The use of relational database principles improves data consistency and efficiency.

Additionally, implementing security measures such as encrypted passwords and access control strengthens the reliability of the system and protects sensitive business information. Future improvements may include automated backups, advanced analytics, and multi-factor authentication for enhanced security.

REFERENCES

1. <https://frappeframework.com/>
2. <https://www.w3schools.com/>
3. <https://owasp.org/>
4. <https://www.php.net/docs.php>